13.4 Mean or Median?

WORKSHEET

The question

STEP II 2000, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q14

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why is the density of the median proportional to $y^n(1-y)^n$? (For Y to be near y, how many of the $2n+1$ values must be below y, and how many above?)

STAGE 1 The constant

Covers: the constant k

Ask yourself A density integrates to 1. Use the integral given in the question with both powers equal to n.

STAGE 2 Mean and variance

Covers: the mean, and the variance

Ask yourself Is the density symmetric about $\frac{1}{2}$? That gives $\mathrm{E}(Y)$.

Pause and try For $\mathrm{E}(Y^2)$, use the given integral again with powers $n+2$ and n, and simplify the factorials.

STAGE 3 Comparing with the mean

Covers: comparing the median with the mean

Ask yourself What are the mean and variance of $\bar{X}$? Compare the two variances for large n.

Pause and try Treat both Y and $\bar{X}$ as normal with mean $\frac{1}{2}$. Write each probability using Φ. Which is larger, and why?

What's really going on

Efficiency

The *relative efficiency* of two unbiased estimators is the ratio of their variances; it says how many times as many observations one needs to match the other. For uniform data the median has efficiency $\frac{1}{3}$ relative to the mean. For normal data it is $\frac{2}{\pi} \approx 0.64$: the median needs about 57% more observations. But for heavy-tailed data the tables turn. For the Cauchy distribution (Section 12.22) the mean is useless, since the average of n observations is no better than one, while the median works well. For the Laplace distribution (Section 12.24) the median is twice as efficient as the mean.

Robust statistics

The median is *robust*: you can change almost half the data to arbitrary values without moving it far. The mean has no such protection: one wrong value can spoil it. In the 1960s and 1970s John Tukey, Peter Huber and others developed *robust statistics*,

estimators that are nearly as efficient as the mean for clean data but resistant to outliers, such as the *trimmed mean* (discard the largest and smallest 10% and average the rest), which is used in judging sports like diving and gymnastics.

Laplace and the median

Laplace showed in 1774 that if errors follow the double exponential (Laplace) distribution, the best estimate of the true value is the median; Gauss showed in 1809 that if they follow the normal distribution, the best estimate is the mean, which led him to the method of least squares. Which is right depends on how errors behave, a question that statisticians still debate for each new kind of data.

Try this next

1. **The median's density.** Derive the density $\frac{(2n+1)!}{(n!)^2} y^n (1-y)^n$ by counting: one value in $[y, y+\delta y]$, n below and n above.

2. **Three observations.** For $n = 1$ (three uniform observations), find $\mathrm{Var}(Y)$ and $\mathrm{Var}(\bar{X})$ exactly. Which is smaller?

3. **The midrange.** Another estimator of the centre is the midrange $\frac{1}{2}(\max + \min)$. Show that for uniform data its variance is $\frac{1}{2(m+1)(m+2)}$ with $m = 2n+1$ observations. How does it compare with the mean? (Compare Section 13.3.)

4. **An outlier.** Eleven measurements are 9.8, 9.9, 10.0, 10.0, 10.1, 10.1, 10.2, 10.2, 10.3, 10.4 and 104 (a typing error). Compare the mean and median.

Further reading

- **John Rice, *Mathematical Statistics and Data Analysis*** (Duxbury), on order statistics and efficiency.
- **Rand Wilcox, *Fundamentals of Modern Statistical Methods*** (Springer). Why robust methods matter, for non-specialists.
- **Stephen Stigler, *The Seven Pillars of Statistical Wisdom*** (Harvard University Press). Aggregation, the mean and the median, in historical perspective.

The full guided solution is in Section 13.4:
https://beyondstep.keithmcnulty.org/ch13/mean-or-median.html